Sampling

Statistical Foundations of AI and ML, Module 4

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Sampling

- Given observations $X = \{x_1, \ldots, x_N\}$, we have learnt to fit a model on them
- Find model f with parameters θ, such that $f_\theta(X)$ is quite high
- But how to generate more observations?
- We need to draw **samples** from f_θ
- The drawn samples should have same statistical properties as original dataset X!
- But how to draw the samples?
- Most basic: Sampling from Uniform Distribution
- Method: Pseudorandom Number Sequence (each number within a range equally likely to occur)

Sampling from Discrete Distributions

- The sampling can start from any point (seed) in the sequence, and proceeds by reading the numbers sequentially
- Scale the drawn number X as $x = a + \frac{(b-a)*(X-min(X))}{max(X)-min(X)}$ where $(min(X), max(X))$ is the range of pseudo-random numbers
- Then $x \sim \mathcal{U}(a, b)$
- This can also be extended to Bernoulli or Categorical Distribution by defining thresholds between (a, b)
- Sample x from $\mathcal{U}(0, 1)$. If $x < p$, let $Y = 1$ else $Y = 0$. Then $Y \sim Bernoulli(p)$
- Simulating N Bernoulli variables like this lets us simulate $Binomial(N, p)$
- $0 < x < p_1 \rightarrow Y = V_1$, $p_1 < x < p_1 + p_2 \rightarrow Y = V_2$, $1 - p_K < x < 1 \rightarrow Y = V_K$. Then $Y \sim Cat(p_1, p_2, \ldots, p_K)$

Sampling from Gaussian

- Sampling from Continuous distributions generally more difficult, but there are tricks
- If $X_1, X_2, \ldots, X_N \sim f_\theta$, then $S_N \sim \mathcal{N}(\mu, \sigma^2/N)$ if $N \to \infty$, and $E(X) = \mu$, $Var(X) = \sigma^2 < \infty$ for any distribution f_θ (where $S_N = 1/N(\sum_{i=1}^{N} X_i)$)
- If $X \sim \mathcal{U}(a, b)$, then $E(X) = (a+b)/2$, $Var(X) = (b-a)^2/12$
- If we can choose (a, b) such that $(a+b)/2 = \mu$ and $(b-a)^2/12 = \sigma^2$, then we can draw a sample from $\mathcal{N}(\mu, \sigma^2)$ by drawing a large number of samples from $\mathcal{U}(a, b)$
- However, this is a time-consuming process! Can we do better?

Sampling from Gaussian

- Solution: Box-Muller Transformation
- Sample $U_1 \sim \mathcal{U}(0,1)$, $U_2 \sim \mathcal{U}(0,1)$
- Now set $R = \sqrt{-2ln(U_1)}$, $\theta = 2\pi U_2$ (polar coordinates)
- $X_1 = Rcos\theta, X_2 = Rsin\theta$
- Now, it can be shown that $X_1 \sim \mathcal{N}(0,1)$, $X_2 \sim \mathcal{N}(0,1)$ (both are independently Gaussian)
- Approach: write the distributions $p(r_1 < R < r_2)$ and $p(s_1 < \theta < s_2)$ separately from the above distributions
- Then calculate $p(a_1 < X_1 < b_1, a_2 < X_2 < b_2)$ using change-of-variables formula

Monte-Carlo Methods

- In many cases, drawing samples allows us to estimate quantities which are difficult to calculate analytically
- Calculation of Expectation: $E_q(f(X)) = \int_{x \in Support(X)} f(x)q(x)dx$ may not be tractable, depending on $f(x)$ and $q(x)$ RV X, follows the distribution 'q'
- Alternative approach, draw N samples $(x_1, x_2, \ldots, x_N)$ from distribution $q(X)$
- Then, $E_q(f(X))$ can be approximated by $\frac{1}{N}\sum_{i=1}^{N} f(x_i)$
- Reason: $\int_{x \in Support(X)} f(x)q(x)dx \approx \sum_{x_i \in Support(X)} f(x_i)q(x_i)$ where x_i are obtained by dividing the interval $Support(X)$ uniformly into many small segments, and the integrand $f(x_i)q(x_i)$ is calculated at each of them
- This is equivalent to choosing x_i with weight proportional to $q(x_i)$, i.e. sampling from $q(X)$

Monte-Carlo Simulation

- Monte-carlo methods are used in many disciplines of science and technology
- In some applications, the aim is to generate new data from a model that has been developed
- In other cases, it is to estimate quantities that are difficult to calculate analytically
- Consider a specific application: estimation of π
- Observation: area of an unit circle$= \pi$, whose limits are $(-1, 1)$ along both dimensions
- Approach: draw samples uniformly from square defined between $(-1, 1)$ (uniformly along both dimensions)
- Count the samples which lie within circle, i.e. $x_i^2 + y_i^2 < 1$, reject the rest
- $\frac{\#pointswithincircle}{\#pointswithinsquare} \approx \frac{areaofcircle}{areaofsquare} = \frac{\pi}{4}$

Rejection Sampling

- I want to draw samples from any PDF f, i.e. choose $x_1, x_2, \dots, x_N$ that have similar statistics as f where $f(x) > 0$, $\int f(x)dx = 1$
- Approach: consider an interval (a, b) such that $a < Supp(f) < b$ and c such that $0 < f(x) < c \forall x \in Supp(f)$
- Rejection Sampling Algorithm:
  - Sample x from $\mathcal{U}(a, b)$ and y from $mathcalU(0, c)$
  - If $y < f(x)$, ACCEPT x else REJECT x
- $prob(xACCEPTED) = \int_{y=0}^{c} prob((x, y)SAMPLED)prob(y < f(x))dy$
- Note: $prob((x, y)SAMPLED) = prob(xSAMPLED) * prob(ySAMPLED) = \frac{1}{c*(b-a)}$
- Also, $prob(y < f(x)) = \frac{f(x)}{c}$
- Multiplying, we find $prob(xACCEPTED) \propto f(x)$, i.e. **the samples follow** f**!**

Rejection Sampling

- Major advantage: Does not need full PDF f, enough to know it upto a constant (unnormalized)
- (Normalizing constant often hard to calculate)
- Can be extended to multi-dimensional distribution by considering high-dimensional bounding box
- If f has unbounded support (eg. $Supp(f) = (-\infty, \infty)$), consider bounding distribution g instead of bounding box
- g should have infinite support (eg. Gaussian), should obey $g(x) > f(x) \forall x$ and g should be easy to sample from
- Problem with Rejection Sampling: many samples may be rejected
- Acceptance probability inversely proportional to height of bounding box c or g

Importance Sampling

- We do not need the full distribution p, but only $E_p(f(X))$
- $E_p(f(X)) = \int_{x \in Supp(X)} f(x)p(x)dx \approx \frac{1}{N} \sum_{i=1}^{N} f(x_i)$ where $x_i \sim p$ and $N \to \infty$
- However, we don't know how to sample from p!
- Consider any arbitrary *proposal distribution* q that is easy to sample from
- $E_p(f(X)) = \int_{x \in Supp(X)} \frac{f(x)p(x)}{q(x)} q(x)dx = E_q(\frac{f(x)p(x)}{q(x)})$
- This can be approximated as $\sum_{i=1}^{N} \frac{f(x_i)p(x_i)}{N q(x_i)}$, where $x_i \sim q$ and $N \to \infty$
- In this case, no sample is rejected
- $\frac{p(x_i)}{q(x_i)}$ are called *importance scores*, because p and q may have different *important* regions

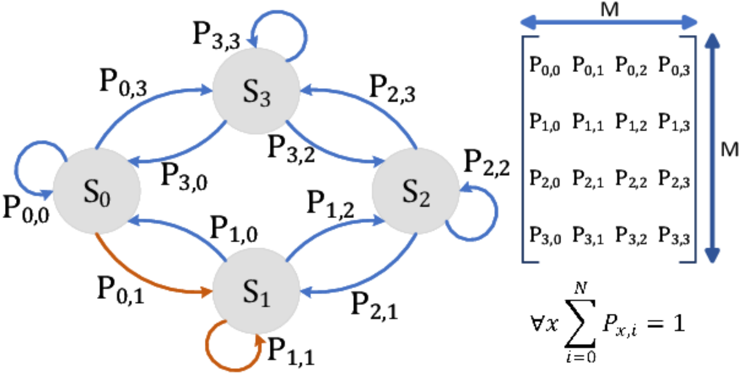
Sampling from irregular distributions

- There may be distributions where the probability mass is concentrated in small pockets of the support set's range
- Example, a Gaussian Mixture where the components have widely different means but very small variances
- Main problem: most samples will be rejected!
- Way out: if one sample is accepted, we would like to search close to it
- *Local Greedy Approach:* always move the samples towards higher density region (analogous to gradient descent)!
- Decide next sample based on current sample (though violates IID sampling principle!)
- Also, beware of multi-modal distributions (may get stuck at single mode only)

Metropolis Algorithm

- At step t, generate proposal x by sampling conditioned on latest sample x^t
- Proposed sample $x \sim g(X|x^t)$, where g may be $\mathcal{N}(x^t, \sigma^2)$ [i.e. proposal is likely to be in the neighborhood of current sample]
- Calculate acceptance ratio as $A(x, x^t) = \frac{f(x)}{f(x^t)}$ where f is the given density
- Draw $u_i \sim \mathcal{U}(0, 1)$
  - If $u_i < A(x, x^t)$, set $x^{t+1} = x$ i.e. ACCEPT and move to new point
  - Else set $x^{t+1} = x^t$ i.e. REJECT and stay in same point
- Note that if f has higher density at new point, we will certainly move there
- Even if it has less density we may move, though with proportionally less chance

Markov Chain



- Markov Chain: set of states with transition probabilities
- Sampling: given current state s, choose next state according to transition distribution P_s
- An aperiodic, recurrent Markov Chain has a stationery distribution over its states

Markov Chain Monte Carlo

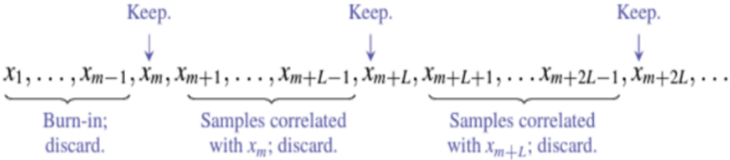
- Do the samples from Metropolis Algorithm represent f?
- Imagine an infinite-state Markov Chain, where each state is a value in the Support Set of f
- An aperiodic, recurrent Markov Chain, with transition distribution Q has a stationery distribution P
- Choose state $x \sim P$, and state $y \sim Q(x)$
- Then, $y \sim P$ (after marginalizing x), i.e. $P(y) = \int_x P(x)Q(y|x)dx$!
- So if we can draw successive samples from a *well-mixed* Markov Chain using the transition distribution, all of them follow the stationery distribution!
- Can we find a Markov Chain with stationary distribution f?
- Yes, if we can define the transition distribution appropriately!

Metropolis-Hastings Algorithm

- We can achieve this by slightly tweaking the acceptance criteria!
- Calculate acceptance ratio as $A(x, x_t) = \frac{f(x)g(x^t|x)}{f(x^t)g(x|x^t)}$ where f is the given density
- Draw $u_i \sim \mathcal{U}(0, 1)$
  - If $u_i < A(x, x^t)$, set $x^{t+1} = x$ i.e. ACCEPT and move to new point
  - Else set $x^{t+1} = x^t$ i.e. REJECT and stay in same point
- Issue 1: How to get x^1? We cannot sample it from f!
- Issue 2: The samples will be correlated, not IID! (identically distributed, but not independent!)
- **Key property** of Markov Chains: start sampling from any arbitrary distribution, but samples will start following stationery distribution after a few steps

Metropolis-Hastings Algorithm

- Let $x^1 \sim \pi$, then $x^2 \sim \pi * Q$, $x^3 \sim \pi * Q * Q$ etc, but $x^t \sim P$ (here * represents marginalization) for $t > t_{mix}$
- The first t_{mix} steps are called *burn-in* or *mixing* phase, but it cannot be counted easily
- Solution to Issue 1: start with any distribution, ignore the first few samples as burn-in (samples not from stationery distribution)
- Solution to Issue 2: do not store all samples, store them at regular intervals
- Interval length can be estimated, such that correlation between stored samples falls off



Gibbs Sampling

- Consider a high-dimensional distribution over $\mathcal{R}^D$
- Sampling new points more difficult in H-D space, most points likely to be rejected
- **Approximation:** move in one dimension at a time!
- Choose a dimension d, and search candidate values as $g(x_d | x_1^t, \ldots, x_{d-1}^t, x_{d+1}^t, \ldots, x_D^t)$
- If sampled value x_d^{new} (obtained from g) is accepted, set $x^{t+1} = \{x_1^t, \ldots, x_{d-1}^t, x_d^{new}, x_{d+1}^t, \ldots, x_D^t\}$
- Repeat above steps for all dimensions in order, over and over
- Eliminate Accept-Reject decision by using f itself as transition distribution g
- Feasible only if $f(x_d | x_1, \ldots, x_{d-1}, x_{d+1}, \ldots, x_D)$ can be calculated

Inference in Latent Variable Models

- Gibbs Sampling is a very useful tool of inference in Latent Variable Models
- Example: Estimation of Z-variables in Hidden Markov Models, $p(Z_t|X)$
- We need $p(Z_t|X, Z_{-t})$ where $Z_{-t} = \{Z_1, \ldots, Z_{t-1}, Z_{t+1}, \ldots, Z_T\}$
- $p(Z_t = k|X, Z_{-t}) \propto p(Z, Z_{-t}, Z_t = k) = p(Z_{t+1}|Z_t = k)p(Z_t = k|Z_{t-1})p(X_t|Z_t = k)$ (using HMM model)
- We know X and have fixed Z_{-t} for now
- $p(Z_{t+1}|Z_t = k)$ and $p(Z_t = k|Z_{t-1})$ can be easily calculated from the state transition distribution
- $p(X_t|Z_t = k)$ can be calculated from emission distribution
- Based on these, we can sample new value of Z_t from $p(Z_t|X, Z_{-t})$

Gibbs Sampling for Latent Variable Models

- General framework of Gibbs Sampling is applicable for all latent variable-based models
- Calculate conditional distributions of each latent variable $p(Z_i|Z_{-i}, X)$
- Initialize all latent variables by any suitable approach (eg. clustering)
- For each iteration
  - Sample each latent variable in order from $p(Z_i|Z_{-i}, X)$ and update it
  - Use the updated values of each latent variables for sampling the next
- If iteration is within burn-in period (say first 50 iterations), just carry on to next iteration
- Else, if iteration is at chosen interval (eg. mod(iteration,10)=0), store the current values of all latent variables

Gibbs Sampling for Latent Variable Models

- Finally, we end up with many samples from each latent variable
- We can now calculate posterior distribution $p(Z_i|X)$ or MAP estimate of each latent variable Z_i
- Burn-in samples are discarded as Markov Chain hasn't mixed yet (we are not sampling from the stationery distribution $P(Z|X)$ as yet)
- After burn-in, sample values stored at regular intervals to avoid correlation between successive draws (we want IID samples)
- Each stored sample basically gives us a snapshot of the latent variables at that point
- There is no concept of convergence (i.e. the latent variables will not become fixed beyond some iteration)
- Rather, the stored values of each random variable will reflect its posterior distribution!